

# ENGINCAN VARAN

Lead AI Engineer | NLP · LLMs · Generative AI

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## PROFESSIONAL SUMMARY

Lead AI Engineer with 6+ years of experience building and deploying production-grade AI systems, specializing in NLP, Speech (STT/TTS), LLMs, and RAG pipelines. Proven track record of architecting end-to-end AI solutions — from model research and fine-tuning to scalable inference and enterprise deployment. Currently leading a team of AI engineers across multiple products, with deep expertise in LLMops, cloud-native AI on OpenShift, and delivering measurable impact across financial and enterprise domains.

## TECHNICAL SKILLS

- **Programming Languages:** Python (*Expert*) · Node.js, JavaScript (*Proficient*) · Java, C++ (*Familiar*)
- **AI & Deep Learning:** PyTorch (*Expert*) · Hugging Face (*Expert*) · TensorFlow-Keras (*Proficient*) · ONNX Runtime, OpenVINO, vLLM (*Proficient*)
- **LLMs & Generative AI:** RAG Pipelines (*Expert*) · Prompt Engineering & LLM Evaluation (*Expert*) · LangChain, LlamaIndex (*Expert*) · LangGraph (*Proficient*) · Agentic AI, Flowise (*Familiar*)
- **NLP & Speech:** STT/TTS (*Expert*) · NER, Text Classification (*Expert*) · Vector DBs — Weaviate, Pinecone, Chroma (*Proficient*) · Reranking, Embedding Models (*Proficient*)
- **Async & Data Systems:** RabbitMQ, Kafka (*Proficient*) · OCR (*Proficient*) · NumPy, Pandas (*Expert*)
- **DevOps & MLOps:** Docker, Kubernetes, OpenShift (*Proficient*) · CI/CD, MLflow, Prometheus, Ragas (*Proficient*) · AWS, Azure (*Familiar*)

## WORK EXPERIENCE

### Lead AI Engineer

TEBArf (TEB Group — AI & Technology), İstanbul, Turkey

May 2026 – Present

- Promoted to Lead AI Engineer, formally taking ownership of team direction, project scoping, and cross-functional stakeholder representation.
- Leading and mentoring a team of **4–5 AI engineers** — conducting technical reviews, defining delivery priorities, and driving engineering quality across all AI products.

### Senior AI Engineer & LLMops Specialist

TEBArf (TEB Group — AI & Technology), İstanbul, Turkey

Jan 2024 – May 2026

- Architecting and deploying enterprise-grade **RAG pipelines** for **8+ enterprise clients** — integrating **RabbitMQ** for async processing, **OCR models** for document ingestion, and advanced chunking strategies with full runtime orchestration.
- Built a custom **RAG and LLM evaluation framework** using **Ragas**, with a purpose-built benchmarking dataset that improved retrieval accuracy by **up to 35%** across domain-specific use cases.
- Defining and enforcing team-wide **engineering standards** — including OOP principles, design patterns (Strategy, Chain of Responsibility, Builder), service routing, API gateway management, and API documentation practices — ensuring maintainable and scalable service architecture across all AI products.
- Designed and maintained production **STT and TTS services** processing **thousands of calls daily**, with both near-real-time and batch capabilities, while continuously improving model accuracy and latency.
- Leading **Intelligent Document Processing (IDP)** for financial and banking documents (contracts, invoices) — combining **LLMs, NLP techniques, and regex-based extraction** to automate information retrieval and document categorization.
- Built and maintained an **AI-powered conversational assistant** integrating STT, TTS, and LLMs for seamless human-machine interaction in production environments.
- Optimized LLM and deep learning inference using **vLLM, OpenVINO, and ONNX Runtime** to reduce latency and improve throughput for production workloads.
- Deployed and maintained **AI services on OpenShift Container Platform**, ensuring high availability, scalability, and reliability across production environments.
- Built an **enterprise complaint classification system** using hierarchical ML (HiClass), achieving **~87% accuracy** and automating routing of **thousands of complaints daily**.

### Software Engineer I & II

Getir, İstanbul, Turkey

Feb 2022 – Jan 2024

- Promoted from SWE I to SWE II within 18 months; led key initiatives including **Order Prioritization** and **Operating Hours Demand Forecasting**.
- Designed and optimized the **GetirWater algorithm** using Python and FastAPI, improving last-mile delivery logistics efficiency by **20–30%**.
- Applied advanced combinatorial optimization (**TSP, VRPTW**) to enhance route planning, reducing operational costs and delivery times across active routes.
- Adapted ML algorithms to industry-specific logistics challenges, delivering measurable improvements in delivery performance.

## Junior Software Engineer

*Yapı Kredi Bank, İstanbul, Turkey*

*Jan 2021 – Jan 2022*

- Contributed to the bank’s first monolithic-to-microservice migration — a pioneering initiative at the institution — delivering full-stack development with **React** and **Java Spring Boot** for core internal banking systems.

## Part-time Engineer & Internship

*Allianz Turkey (Part-time) · Akbank (Intern), İstanbul, Turkey*

*Jun 2020 – Jan 2021*

- Modernized web applications with **Angular, Ionic, and Java** at Allianz; built back-end services with **.NET and Java** at Akbank — gaining early exposure to enterprise-scale production systems.

## EDUCATION

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### M.Sc. Computer Science (All Coursework Completed)

*Sabancı University, İstanbul, Turkey*

*2021 – 2024*

- cGPA: 4.0 · Completed graduate-level coursework in AI, deep learning, and cybersecurity
- Research focus: Deep learning approaches for detecting control flow-based attacks; work contributed to a peer-reviewed publication

### B.Sc. Computer Science

*Sabancı University, İstanbul, Turkey*

*2017 – 2021*

- cGPA: 3.8
- Mathematics Minor Degree

### Computer Science (Erasmus Exchange)

*Mälardalens Högskola, Västerås, Sweden*

*2019 – 2020*

### High School

*TEV İnanç Türkeş High School, Kocaeli, Turkey*

*2012 – 2017*

- Selective gifted & talented institution admitting students based on national academic merit exams

## PUBLICATIONS

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- Engincan Varan, “Using Page Offsets for Detecting Control-Flow Anomalies,” *Innovative Security Solutions for Information Technology and Communications*, Springer, 2023. [DOI: 10.1007/978-3-031-52947-4\_2]

## LANGUAGES AND HOBBIES

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### Languages Known

- **Turkish:** Native
- **English:** Advanced (C2)
- **German:** Basic (A2)

### Hobbies

- 🏸 Tennis (Part of TEB&TEBARf Tennis Team)
- 🥁 Drumming
- 🏹 Archery